

UNIVERSIDAD MARIANO GÁLVEZ DE GUATEMALA

FACULTAD: Ingeniería En Sistemas

SEDE: Chimaltenango

CARRERA: Ingeniería En Sistemas De La Información y Ciencias De La Computación

CURSO: Sistemas Operativos I

CATEDRÁTICO: Josué Giménez

TEMA: Sprint 2



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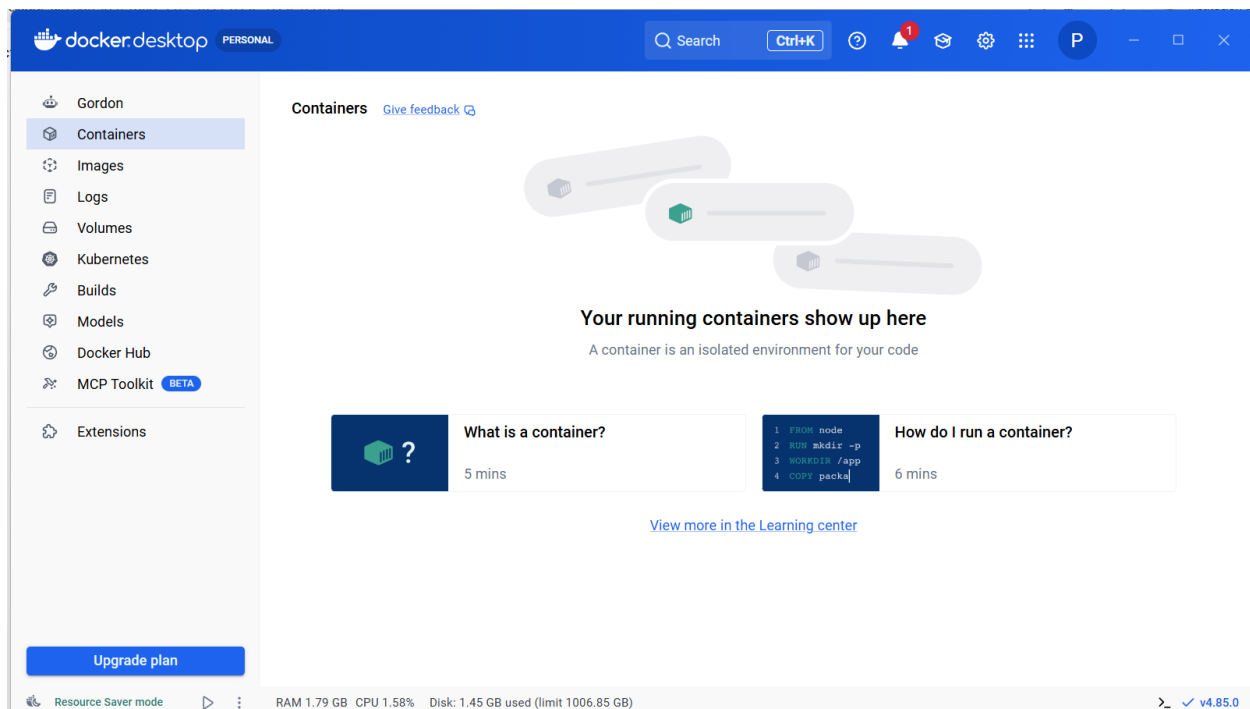
Chimaltenango 05 de agosto de 2026

Captura 1:

Descarga e instalación de Docker Desktop y verificando que este correctamente funcionando desde el PowerShell

```
Administrador: Windows PowerShell
Windows PowerShell 5.1
Copyright (C) Microsoft Corporation. Todos los derechos reservados.

PS C:\WINDOWS\system32> wsl --version
>> docker --version
>> docker ps
Versión de WSL: 2.7.10.0
Versión de kernel: 6.18.33.2-2
Versión de WSLg: 1.0.73.2
Versión de MSRDC: 1.2.6676
Versión de Direct3D: 1.611.1-81528511
Versión de DXCore: 10.0.26100.1-240331-1435.ge-release
Versión de Windows: 10.0.26200.8737
Docker version 29.6.2, build dfc4efb
failed to connect to the docker API at npipe://./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
PS C:\WINDOWS\system32>
```



Captura 1.1:

Comando Artisan personalizado (so:monitor) mostrando en tiempo real el uso de CPU, memoria y disco del sistema.

```
pichiya@DESKTOP-597FKA9: x + v

$usedKb = $totalKb - $availableKb;

$percent = $totalKb > 0 ? ($usedKb / $totalKb) * 100 : 0;
$label = sprintf('%1fGB / %1fGB', $usedKb / 1048576, $totalKb / 1048576);

return ['percent' => round($percent, 1), 'label' => $label];
}

protected function getDiskUsage()
{
    $total = disk_total_space('/');
    $free = disk_free_space('/');
    $used = $total - $free;

    $percent = $total > 0 ? ($used / $total) * 100 : 0;
    $label = sprintf('%1fGB / %1fGB', $used / 1073741824, $total / 1073741824);

    return ['percent' => round($percent, 1), 'label' => $label];
}

pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ php artisan so:monitor --once
Iniciando monitoreo del Sistema Operativo...
Presiona CTRL+C para detener.

[2026-08-05 20:48:39]
CPU [.....] 0.8%
Memoria [.....] 2.6% (0.8GB / 31.0GB)
Disco [.....] 5.3% (53.8GB / 1006.9GB)

pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ |
```

Ingresamos desde nuestras carpetas de laravel y las carpetas de mi proyecto para inicializar el Sprint 2

```
pichiya@DESKTOP-597FKA9: x + v

total 396
drwxr-xr-x 12 pichiya pichiya 4096 Jul 26 20:28 .
drwxr-xr-x  4 pichiya pichiya 4096 Jul 26 21:06 ..
-rw-r--r--  1 pichiya pichiya 275 May 25 17:11 .editorconfig
-rw-r--r--  1 pichiya pichiya 1137 Jul 26 20:28 .env
-rw-r--r--  1 pichiya pichiya 1086 May 25 17:11 .env.example
-rw-r--r--  1 pichiya pichiya 186 May 25 17:11 .gitattributes
-rw-r--r--  1 pichiya pichiya 341 May 25 17:11 .gitignore
-rw-r--r--  1 pichiya pichiya 31 May 25 17:11 .npmrc
-rw-r--r--  1 pichiya pichiya 3708 May 25 17:11 README.md
drwxr-xr-x  5 pichiya pichiya 4096 May 25 17:11 app
drwxr-xr-x  1 pichiya pichiya 425 May 25 17:11 artisan
drwxr-xr-x  3 pichiya pichiya 4096 May 25 17:11 bootstrap
-rw-r--r--  1 pichiya pichiya 2883 May 25 17:11 composer.json
-rw-r--r--  1 pichiya pichiya 306108 Jul 26 20:28 composer.lock
drwxr-xr-x  2 pichiya pichiya 4096 May 25 17:11 config
drwxr-xr-x  5 pichiya pichiya 4096 Aug 1 11:42 database
-rw-r--r--  1 pichiya pichiya 384 May 25 17:11 package.json
-rw-r--r--  1 pichiya pichiya 1322 May 25 17:11 phpunit.xml
drwxr-xr-x  2 pichiya pichiya 4096 May 25 17:11 public
drwxr-xr-x  5 pichiya pichiya 4096 May 25 17:11 resources
drwxr-xr-x  2 pichiya pichiya 4096 May 25 17:11 routes
drwxr-xr-x  4 pichiya pichiya 4096 May 25 17:11 storage
drwxr-xr-x  4 pichiya pichiya 4096 May 25 17:11 tests
drwxr-xr-x 39 pichiya pichiya 4096 Jul 26 20:28 vendor
-rw-r--r--  1 pichiya pichiya 632 May 25 17:11 vite.config.js
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel$ cd app
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ php artisan --version
Laravel Framework 13.22.0
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ |

Microsoft Windows [Versión 10.0.26200.8737]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\Pichiya>wsl
pichiya@DESKTOP-597FKA9:/mnt/c/Users/Pichiya$ cd ~/proyectos/app-so-laravel
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel$ ls -la
total 36
drwxr-xr-x  4 pichiya pichiya 4096 Jul 26 21:06 .
drwxr-xr-x  3 pichiya pichiya 4096 Jul 26 19:05 ..
drwxr-xr-x  7 pichiya pichiya 4096 Jul 26 21:31 .git
-rw-r--r--  1 pichiya pichiya 581 Jul 26 21:07 .gitignore
-rw-r--r--  1 pichiya pichiya 1121 Jul 26 21:06 Dockerfile
-rw-r--r--  1 pichiya pichiya 53 Jul 26 19:05 README.md
drwxr-xr-x 12 pichiya pichiya 4096 Jul 26 20:28 app
-rw-r--r--  1 pichiya pichiya 1527 Jul 26 21:06 docker-compose.yml
-rw-r--r--  1 pichiya pichiya 1044 Jul 26 21:06 supervisord.conf
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  app/app/Console/

nothing added to commit but untracked files present (use "git add" to track)
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ mkdir -p app/app/Console/Commands
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ touch app/app/Console/Commands/MonitorSO.php
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ cd app
pichiya@DESKTOP-597FKA9:~/proyectos/app-so-laravel/app$ php -l app/Console/Commands/MonitorSO.php
No syntax errors detected in app/Console/Commands/MonitorSO.php
```

Captura 2:

Verificacion de los 3 contenedores Docker corriendo (laravel-app, laravel-mysql, laravel-redis) mediante docker ps.

```
pichiya@DESKTOP-597FKA9: ~ - el-app Built
pichiya@DESKTOP-597FKA9: ~/projectos/app-so-laravel$ docker-compose down
WARN[0000] /home/pichiya/projectos/app-so-laravel/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] down 4/4
  ✓ Container laravel-app      Removed          11.3s
  ✓ Container laravel-mysql    Removed          2.0s
  ✓ Container laravel-redis    Removed          0.6s
  ✓ Network app-so-laravel_laravel-net Removed          0.3s
WARN[0000] /home/pichiya/projectos/app-so-laravel/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] up 4/4
  ✓ Network app-so-laravel_laravel-net Created          0.1s
  ✓ Container laravel-redis     Started           0.8s
  ✓ Container laravel-mysql     Started           0.8s
  ✓ Container laravel-app       Started           1.0s
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ docker ps
docker exec laravel-app supervisorctl -c /etc/supervisor/conf.d/supervisord.conf status
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS
8972032ad7e   app-so-laravel-app "docker-php-entrypoint..." About a minute ago Up About a minute (unhealthy) 0.0.0.
0:8000->8000/tcp, [::]:8000->8000/tcp laravel-app
7e7ec8c6057f   redis:7-alpine   "docker-entrypoint.s..." About a minute ago Up About a minute (healthy) 0.0.0.
0:6379->6379/tcp, [::]:6379->6379/tcp laravel-redis
5c90eb5ecef6   mysql:8.0        "docker-entrypoint.s..." About a minute ago Up About a minute (healthy) 0.0.0.
0:3306->3306/tcp, [::]:3306->3306/tcp laravel-mysql
laravel-services:laravel-app   RUNNING   pid 7, uptime 0:01:43
laravel-services:laravel-monitor FATAL     Exited too quickly (process log may have details)
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$
```

```
pichiya@DESKTOP-597FKA9: ~ -
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ cd app
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel/app$ php -l app/Console/Commands/MonitorS0.php
No syntax errors detected in app/Console/Commands/MonitorS0.php
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel/app$ cd ~/projectos/app-so-laravel
pwd
/home/pichiya/projectos/app-so-laravel
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ docker-compose build

The command 'docker-compose' could not be found in this WSL 2 distro.
We recommend to activate the WSL integration in Docker Desktop settings.

For details about using Docker Desktop with WSL 2, visit:
https://docs.docker.com/go/wsl2/

pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ wsl
cd ~/projectos/app-so-laravel
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ docker-compose --version
Docker Compose version v5.3.1
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ docker-compose build
WARN[0000] /home/pichiya/projectos/app-so-laravel/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Building 0.0s (0/0)
[*] build 0/1
  " Image app-so-laravel-app Building
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
pichiya@DESKTOP-597FKA9:~/projectos/app-so-laravel$ ex
```

```
pichiya@DESKTOP-597FKA9: ~ -
C:\Users\Pichiya>wsl
pichiya@DESKTOP-597FKA9: /mnt/c/Users/Pichiya$ groups
pichiya adm cdrom sudo dip plugdev users docker
pichiya@DESKTOP-597FKA9: /mnt/c/Users/Pichiya$ cd ~/projectos/app-so-laravel
docker-compose build
WARN[0000] /home/pichiya/projectos/app-so-laravel/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[*] Building 87.7s (13/17)
=> [internal] load local bake definitions          0.0s
=> => reading from stdin 530B                      0.0s
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 1.16kB                0.0s
=> [internal] load metadata for docker.io/library/php:8.2-fpm 3.4s
=> [auth] library/php:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                   0.1s
=> => transferring context: 2B                        0.0s
=> [ 1/11] FROM docker.io/library/php:8.2-fpm@sha256:91f09dc9fcc906fa1198d64a2f50e51b69f420942ded7a3cb7707fb51b 29.7s
=> => resolve docker.io/library/php:8.2-fpm@sha256:91f09dc9fcc906fa1198d64a2f50e51b69f420942ded7a3cb7707fb51b013 0.0s
=> sha256:a8342b2036d3ec1561d68414df7d7b61ea22e0a1742b726a7e9c18201e3661 9.25kB / 9.25kB 0.2s
=> sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0c0b5577484a6d75e68dc38e8acc1 32B / 32B 0.3s
=> sha256:4d1be1c8b07fd1ea2d3d8696bdc83d8ac874081f924c60db26294503d9857250 245B / 245B 0.4s
=> sha256:c79914e7231d8b243d3d31629aee1afbab20549276975b307c39fe9a33f6fb41 251B / 251B 0.5s
=> sha256:4d1be1c8b07fd1ea2d3d8696bdc83d8ac874081f924c60db26294503d9857250 2.45kB / 2.45kB 0.3s
=> sha256:44e01373f679de21da7d3e30e644f76ae6f2c8b4fc20f95da75947f8ca4c8f37 487B / 487B 0.2s
=> sha256:0615565e01d52ec689c939e69fb1bc61e3657850274c161c72bcf1ef90985b8 11.65MB / 11.65MB 2.7s
=> sha256:28c422a84b5ba04fafecf3a0966fbc16474d9628d2c2999b764c390b0c3b91c 12.31MB / 12.31MB 4.2s
=> sha256:1f6e7aa22ef01aaff855c206819f4ad3068ed571e1b5ab7cb4906610bce0f17d 224B / 224B 0.2s
=> sha256:49c8e0356295cbd3852c50f409cf637625789f0efd94f4aa90199b068c688c3 117.84MB / 117.84MB 22.5s
=> sha256:176298f15cbcf592a89b616bedf0ddcb72d917259a07fe105578ae87f3069d14 226B / 226B 0.3s
=> sha256:26c307b5e35a59ce911f5fde5b9458120ec8734e831ea2da5649a9ad14abfd3d 29.78MB / 29.78MB 19.3s
```

Captura 3:

Monitoreo en vivo del comando `so:monitor` ejecutandose dentro del contenedor Docker (`docker exec`).

```
pichiya@DESKTOP-597FKA9: ~
CPU [.....] 1.7%
Memoria [#.....] 6.1% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)

[2026-08-05 22:35:24]
CPU [.....] 1.6%
Memoria [#.....] 6.1% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)

[2026-08-05 22:35:29]
CPU [.....] 1.5%
Memoria [#.....] 6.0% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)

[2026-08-05 22:35:34]
CPU [.....] 1.4%
Memoria [#.....] 6.0% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)

[2026-08-05 22:35:39]
CPU [.....] 1.2%
Memoria [#.....] 6.0% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)

[2026-08-05 22:35:44]
CPU [.....] 1.1%
Memoria [#.....] 6.0% (1.9GB / 31.0GB)
Disco [#.....] 5.5% (55.8GB / 1006.9GB)
```

Captura 4:

Supervisor gestionando ambos procesos (`laravel-app` y `laravel-monitor`) como daemons en estado `RUNNING` (`supervisorctl status`).

```
at vendor/monolog/monolog/src/Monolog/Handler/StreamHandler.php:164
160     }
161     if (!is_resource($stream)) {
162         $this->stream = null;
163     }
→ 164     throw new \UnexpectedValueException(sprintf('The stream or file "%s" could not be opened in app
end mode: '.$this->errorMessage, $url) . Utils::getRecordMessageForException($record));
165     }
166     stream_set_chunk_size($stream, $this->streamChunkSize);
167     $this->stream = $stream;
168     $this->inodeUrl = $this->getInodeFromUrl();

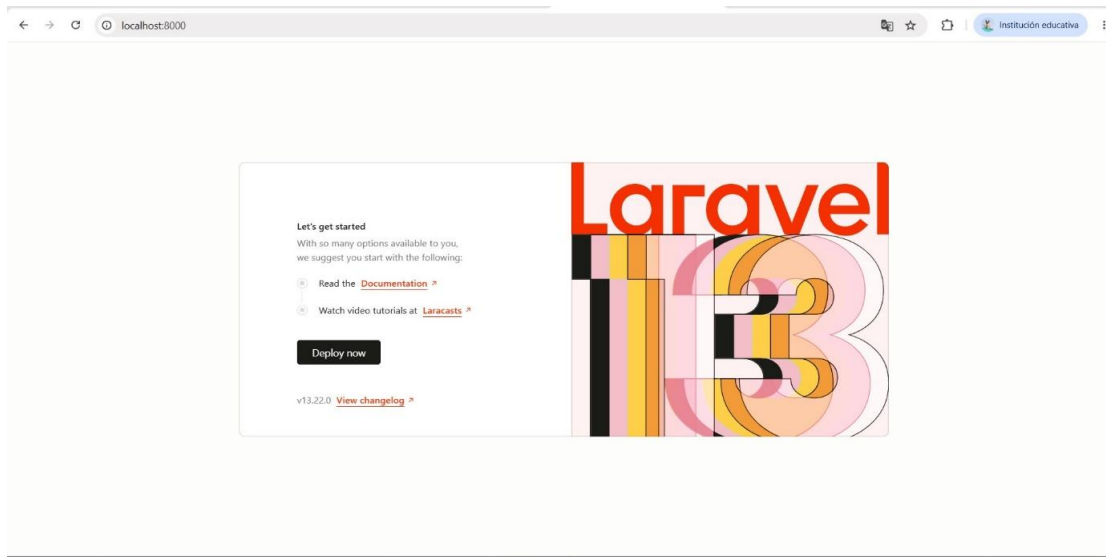
+12 vendor frames

13 artisan:16
Illuminate\Foundation\Application::handleCommand(Object(Symfony\Component\Console\Input\ArgvInput))

pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ docker exec laravel-app chown -R www-data:www-data /app/storage
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ docker exec laravel-app chmod -R 775 /app/storage
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ docker exec laravel-app supervisorctl -c /etc/supervisor/conf.d/supe
rvisor.conf restart laravel-services:laravel-monitor
laravel-services:laravel-monitor: ERROR (not running)
laravel-services:laravel-monitor: started
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ ^C
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ docker exec laravel-app supervisorctl -c /etc/supervisor/conf.d/supe
rvisor.conf status
laravel-services:laravel-app      RUNNING   pid 7, uptime 0:04:50
laravel-services:laravel-monitor  RUNNING   pid 157, uptime 0:00:14
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$
```

Captura 5:

Acceso web exitoso a la aplicacion Laravel en <http://localhost:8000>, sirviendo la pagina de bienvenida con conexion a MySQL.



Captura 6

Script Monitor-Laravel.ps1 ejecutado desde PowerShell (Windows), mostrando el estado de los daemons, ultimas lecturas del log y verificacion de acceso web.

```
Administrador: Windows PowerShell
PS Microsoft.PowerShell.Core\FileSystem::\\wsl$\Ubuntu\home\pichiya\proyectos\app-so-laravel> .\Monitor-Laravel.ps1
=====
MONITOR LARAVEL - SPRINT 2 (Sistemas Operativos)
=====

Contenedor 'laravel-app' esta corriendo.

--- Estado de los daemons (Supervisor) ---
laravel-services:laravel-app      RUNNING    pid 7, uptime 0:14:17
laravel-services:laravel-monitor  RUNNING    pid 8, uptime 0:14:17

--- Ultimas lecturas de monitor.log ---
CPU      [.....] 0.6%
Memoria  [#.....] 6.0% (1.9GB / 31.0GB)
Disco    [#.....] 5.6% (55.9GB / 1006.9GB)

[2026-08-05 23:58:06] CPU: 0.6% | Memoria: 6.0% (1.9GB / 31.0GB) | Disco: 5.6% (55.9GB / 1006.9GB)
[2026-08-05 23:58:06]
CPU      [.....] 0.6%
Memoria  [#.....] 6.0% (1.9GB / 31.0GB)
Disco    [#.....] 5.6% (55.9GB / 1006.9GB)

--- Verificando acceso web (http://localhost:8000) ---
Servidor respondio con codigo: 200

=====
Monitoreo completado - 2026-08-05 17:58:08
=====
PS Microsoft.PowerShell.Core\FileSystem::\\wsl$\Ubuntu\home\pichiya\proyectos\app-so-laravel>
```

Captura 7:

Commit de Git con todos los cambios del Sprint 2: comando so:monitor, configuracion de Docker/Supervisor y script de PowerShell.

```
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ docker-compose down
WARN[0000] /home/pichiya/proyectos/app-so-laravel/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ cd ~/proyectos/app-so-laravel
git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Dockerfile
    modified:   app/storage/app/.gitignore
    modified:   app/storage/app/private/.gitignore
    modified:   app/storage/app/public/.gitignore
    modified:   app/storage/framework/.gitignore
    modified:   app/storage/framework/cache/.gitignore
    modified:   app/storage/framework/cache/data/.gitignore
    modified:   app/storage/framework/sessions/.gitignore
    modified:   app/storage/framework/testing/.gitignore
    modified:   app/storage/framework/views/.gitignore
    modified:   docker-compose.yml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Monitor-Laravel.ps1
    app/app/Console/

no changes added to commit (use "git add" and/or "git commit -a")
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ |
```

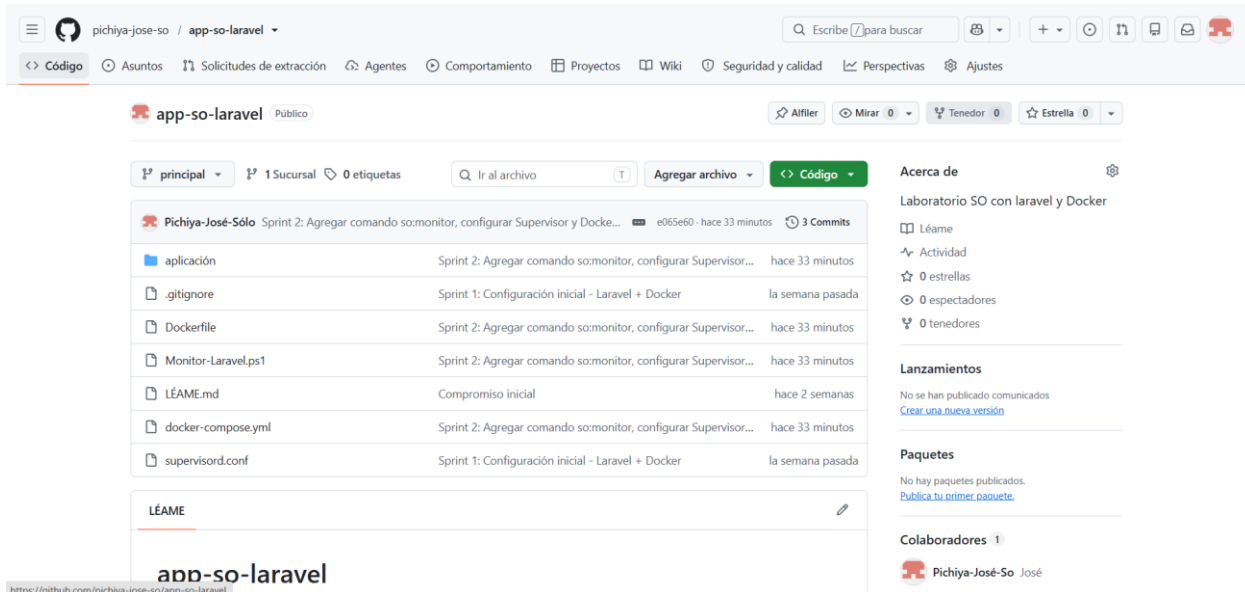
Captura 8:

Push exitoso de los cambios al repositorio remoto en GitHub.

```
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ git push origin main
Username for 'https://github.com': jose
Password for 'https://jose@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/pichiya-jose-so/app-so-laravel.git/'
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ jose123
jose123: command not found
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ git push origin main
Username for 'https://github.com': jose
Password for 'https://jose@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/pichiya-jose-so/app-so-laravel.git/'
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ git push origin main
Username for 'https://github.com': ghp_K5KZwuUfIjJANoCDDLXMBt4met43n0ubL
Password for 'https://ghp_K5KZwuUfIjJANoCDDLXMBt4met43n0ubL@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/pichiya-jose-so/app-so-laravel.git/'
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ git push origin main
Username for 'https://github.com': jose
Password for 'https://jose@github.com':
Enumerating objects: 25, done.
Counting objects: 100% (25/25), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (15/15), 3.33 KiB | 567.00 KiB/s, done.
Total 15 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/pichiya-jose-so/app-so-laravel.git
   df2d942..e065e60  main -> main
pichiya@DESKTOP-597FKA9: ~/proyectos/app-so-laravel$ |
```

Captura 9:

Se cargaron las carpetas del Sprint 2 al repositorio Github.com quedando en funcionamiento



ghp_KDJUzHbV5wgZzbQ1Ab4DD3sqJEjZFd0OKgJE

ghp_K5KZwuUfIjJANoCDDLXMBtTaxT4met43n0ubL

URL para ingreso a Github.com

<https://github.com/pichiya-jose-so/app-so-laravel>

Conclusión

Al realizar la practica tuvo como finalidad la implementación de un sistema de monitoreo de recursos de sistema operativos los cuales son CPU, RAM y disco duro integrado a la aplicación laravel conectada con docker y gestionada mediante supervisor como administrador de proceso daemon, en el desarrollo del mismo permite comprender de forma práctica como un daemon a diferencia de un proceso convencional que se ejecuta una vez y termina se mantiene corriendo de manera continua en segundo plano, reiniciándose automáticamente ante fallos de tales como lo demostró supervisor al gestionar simultáneamente el servidor web laravel y el comando de monitoreo dentro del mismo conector.